

fvfmPy and fvfmR

User's manual

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Basic Operation

File preparation

fvfmPy reads fluorescence files from a Walz imaging fluorometer in either .pim format (proprietary Walz format) or multi-frame .tif file. The multi-frame .tif format can be supplied from the Walz ImagingWin software. Files to be analyzed should be collected into a common folder with unique names. You may combine .tif and .pim files in one folder; there is no need to specify file type.

Running the GUI

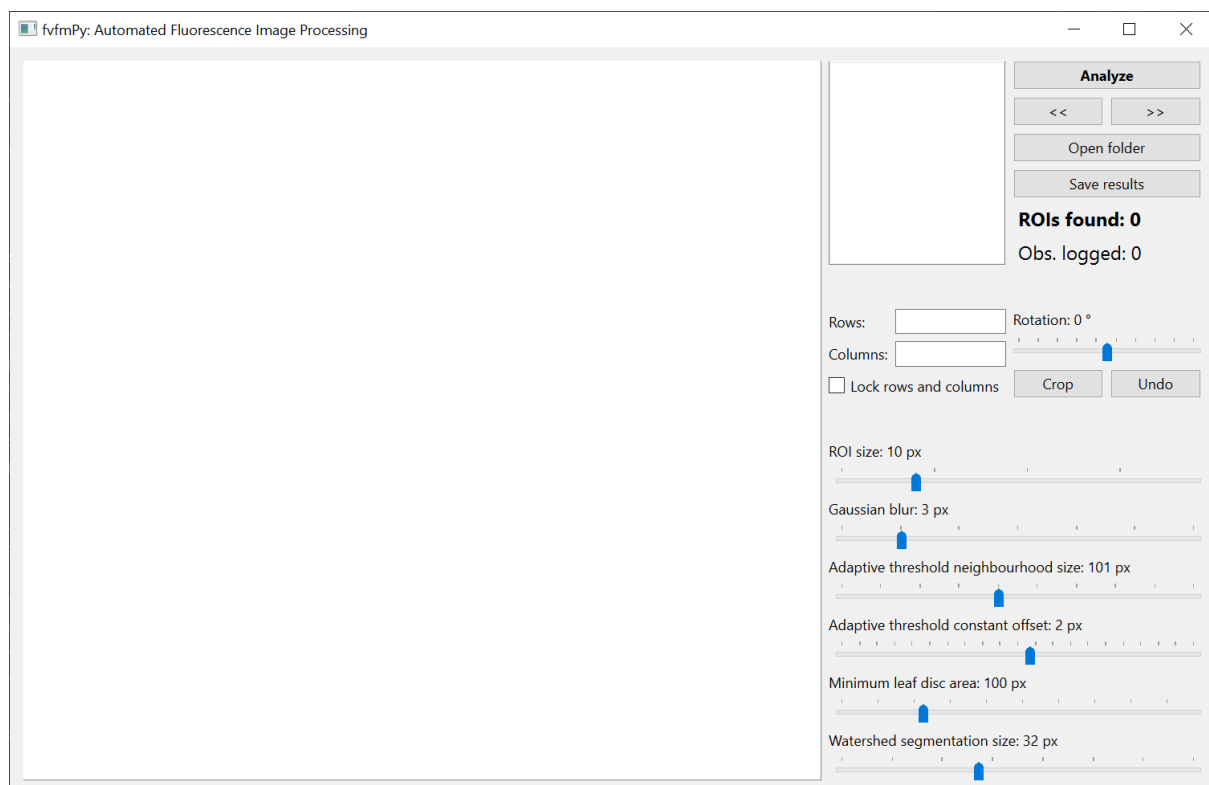


Figure 1 - Graphical user interface.

fvfmPy may be run from the command line or using an integrated development environment. Alternatively, fvfmPy may be run using the R wrapper, fvfmR. Once run, the user will be presented with the graphical user interface (GUI) shown in Fig. 1.

Opening a folder

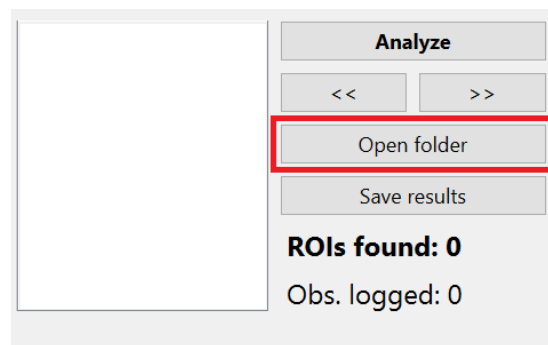


Figure 2 - Opening a folder.

To begin analysis, the user will first open a folder containing .pim or .tif files by clicking the button shown in Fig. 2.

Inspecting and adjusting ROIs

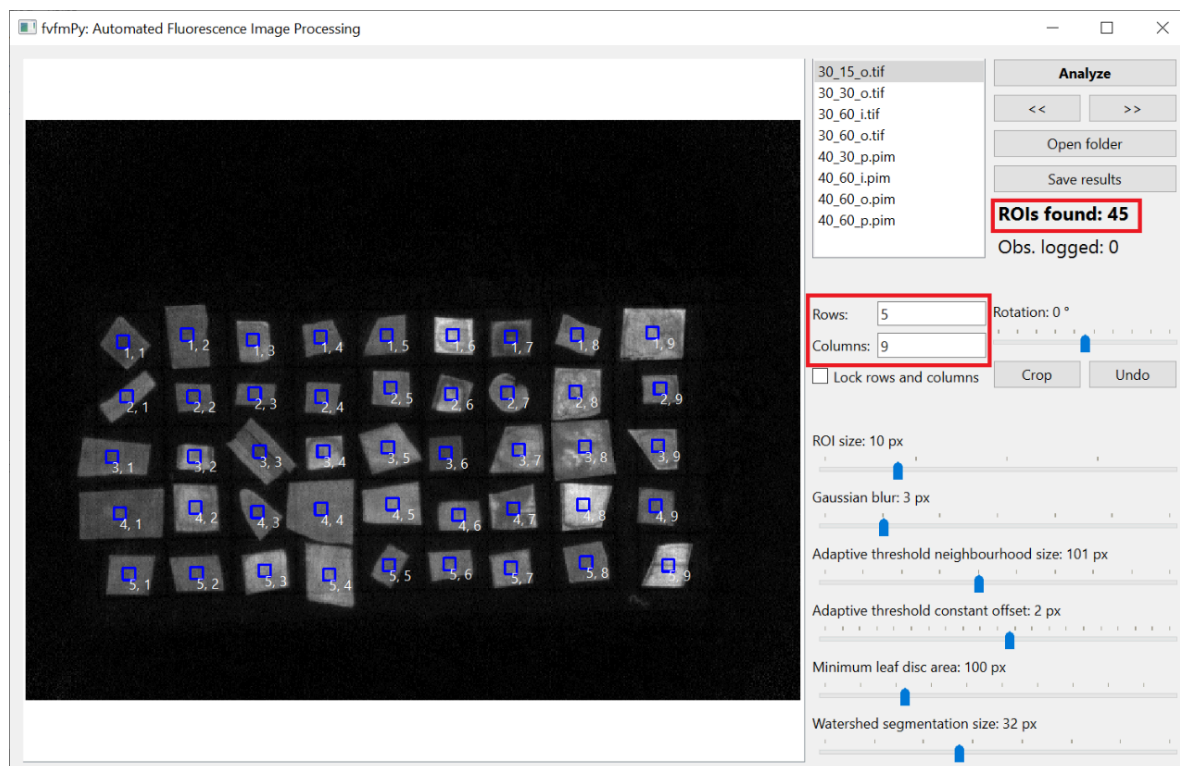


Figure 3 - Automated image analysis and ROI selection. Number of ROIs detected and estimated number of rows and columns are highlighted in red.

Opening a folder will immediately load the first image into the viewer window as shown in Fig. 3. The software will analyze the image and attempt to estimate the best locations for regions of interest (ROIs). The number of ROIs found in the current image is displayed in the top-right of the GUI. The software also attempts to estimate the number of rows and columns in the array, which is also displayed in the GUI. **If the ROI assignments are not correct,**

the user may adjust them by double-clicking on an existing ROI to remove it, or double-clicking an empty region to add an ROI.

Analyzing the image

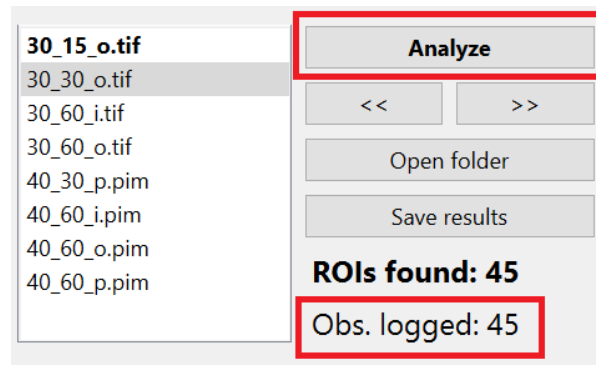


Figure 4 - Analyzing ROIs and extracting fluorescence metrics. Analyze button and number of logged observations are highlighted.

Once the user is satisfied with the placement of the ROIs, they may **analyze the ROIs and extract fluorescence metrics for the current image by pressing the “Analyze” button, or by pressing the Enter key on their keyboard.** This logs the current fluorescence metrics and immediately advances to the next image in the folder. Completed files are highlighted in bold in the file viewer. The total number of observations logged in this way are displayed in the GUI.

Navigation and reanalysis

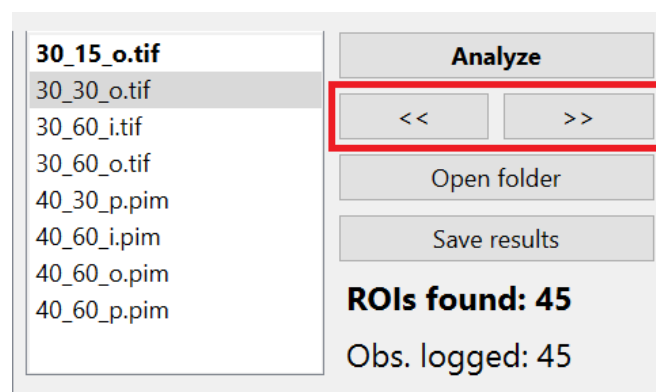


Figure 5 - Navigation. Skip forward and backward buttons are highlighted

The user may navigate backward or forward in their current folder by pressing the “<<” or “>>” buttons, respectively, as highlighted in Fig. 5. If the user wishes to reanalyze an image that they previously analyzed, they may do so by navigating to the image in question and pressing the “Analyze” button or Enter key. **Reanalyzing the same image will overwrite the previous data logged for that file;** the user will be warned of this by a dialog box.

Saving data

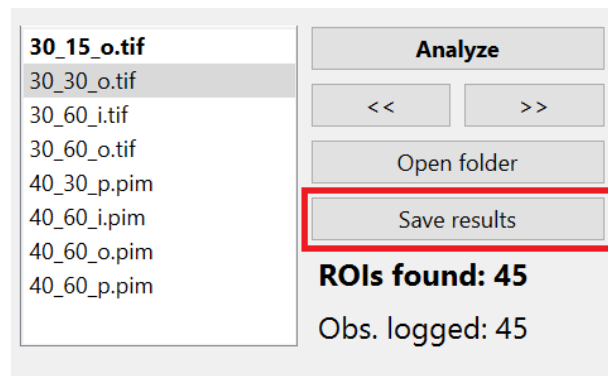


Figure 6 - Save results button.

Extracted fluorescence metrics are stored in working memory and must be saved by the user. **When the user wishes to save their data, they may press the “Save results” button**, as highlighted in Fig. 6. The user will then have the option to select a destination file for their results. The results are stored in a plain-text comma-separated value (.csv) file format. The output file format is described in Table 1.

Table 1: Output file format metadata

Column name	Description	Units or datatype
filename	Name of processed fluorescence datafile (.pim or .tif)	String
row	Row number of leaf disc array	Integer
col	Column number of leaf disc array	Integer
leaf_number	Unique number of leaf disc array within file	Integer
centroid_x	X-coordinate of centroid of ROI	Pixels
centroid_y	Y-coordinate of centroid of ROI	Pixels
mean_Fo	Mean Fo within ROI	Uncalibrated fluorescence value (unitless)
mean_Fm	Mean Fm within ROI	Uncalibrated fluorescence value (unitless)
FvFm	Mean Fv/Fm within ROI	Ratio (unitless)

Optional processing steps

We provide the user with additional options for processing their images which can improve the performance of the image segmentation and ROI selection algorithms in some cases. Note that many of these additional processing steps will result in the loss of previous manual adjustments to ROIs performed on the current image. Therefore, **we recommend that the user perform these optional steps first (if desired) before making any final manual adjustments to ROI placement.**

Rotate and crop

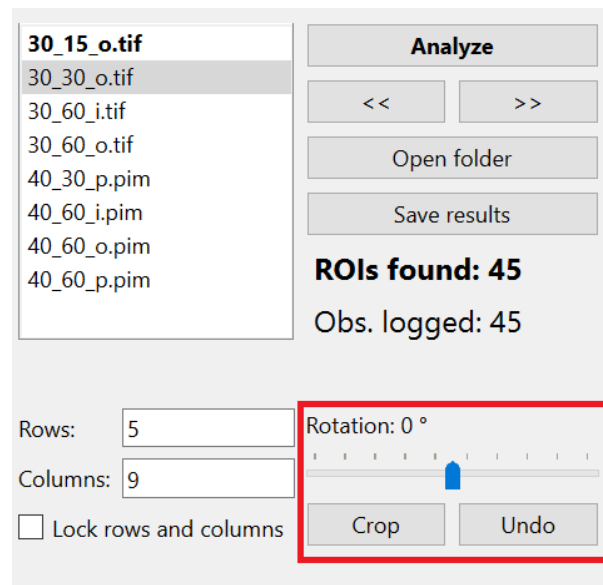


Figure 7 - Rotate and crop controls.

Occasionally, the user may want to rotate or crop their image to improve segmentation performance. Rotation may affect row and column assignment, and presence of debris or a bright background may warrant cropping of unwanted image area. Rotation and cropping controls are highlighted in Fig. 7.

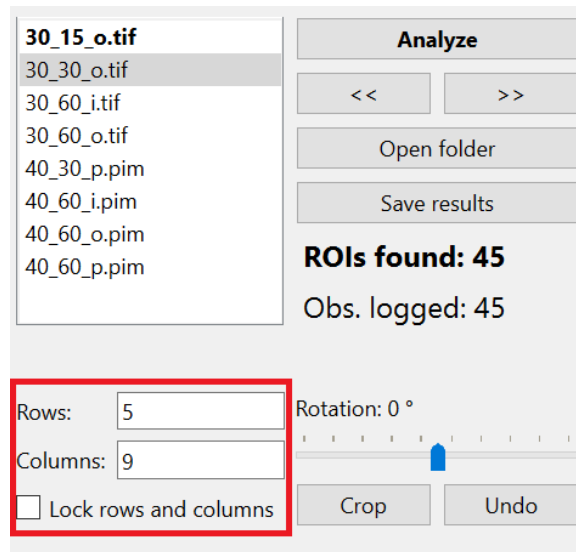
To rotate the image, the user drags the “Rotation” slider to the left or right until their desired level of rotation is reached. As the image is rotated, the software will continuously attempt to perform ROI identification.

To crop the image, the user presses the “Crop” button, then clicks and drags the mouse over the desired area in the image viewer. When the mouse button is released, the image is cropped to the selected area and ROI identification is performed on the cropped image. The user may revert to the original image at any time by pressing the “Undo” button.

Note 1: Cropping the image will lock the current rotation in place. If the user wishes to rotate the image after cropping, they must press “Undo” to revert to the original image, perform their desired rotation, and then re-crop the image.

Note 2: Cropping or rotating the image will result in loss of manual adjustments to ROIs.

Row and column specification



The screenshot shows a software interface with a file list on the left containing files like 30_15_o.tif, 30_30_o.tif, 30_60_i.tif, 30_60_o.tif, 40_30_p.pim, 40_60_i.pim, 40_60_o.pim, and 40_60_p.pim. On the right, there are buttons for 'Analyze', '<<', '>>', 'Open folder', and 'Save results'. Below these, it displays 'ROIs found: 45' and 'Obs. logged: 45'. At the bottom, there are input fields for 'Rows: 5' and 'Columns: 9', a 'Rotation: 0 °' slider, a 'Lock rows and columns' checkbox, and 'Crop' and 'Undo' buttons. A red rectangle highlights the 'Rows' and 'Columns' input fields.

Figure 8 - Row and column specification.

The software automatically estimates the number of rows and columns in the current array and assigns each ROI to a row and column. **If this estimate is incorrect, the user may manually override this by changing the number of rows and columns in the boxes highlighted in Fig. 8.** The user also has the option to lock the current number of rows and columns by checking the indicated box. This will prevent the software from trying to estimate the number of rows and columns in each image, and it will instead use the values provided.

Note 1: Changing the row and column specification will change the assignment of each identified ROI into the specified number of rows and columns, but does not affect the identification of ROIs themselves.

Note 2: If the user does not lock the row and column numbers, then any further image processing steps (e.g., adjusting parameters; next section) will result in the software re-attempting to estimate the row and column values.

Image segmentation parameter adjustment

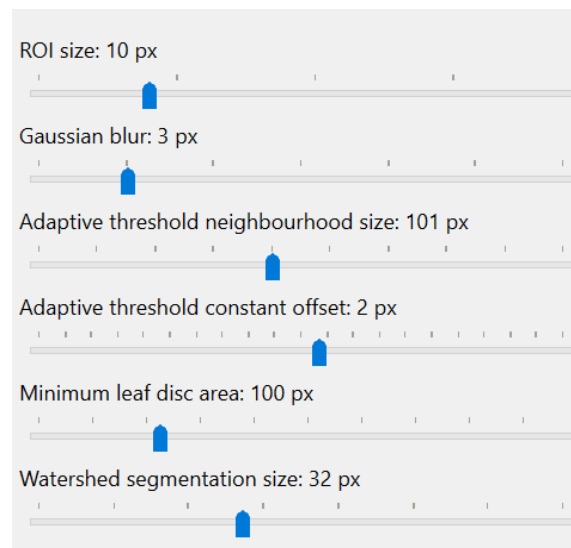


Figure 9 - Image segmentation parameter adjustments.

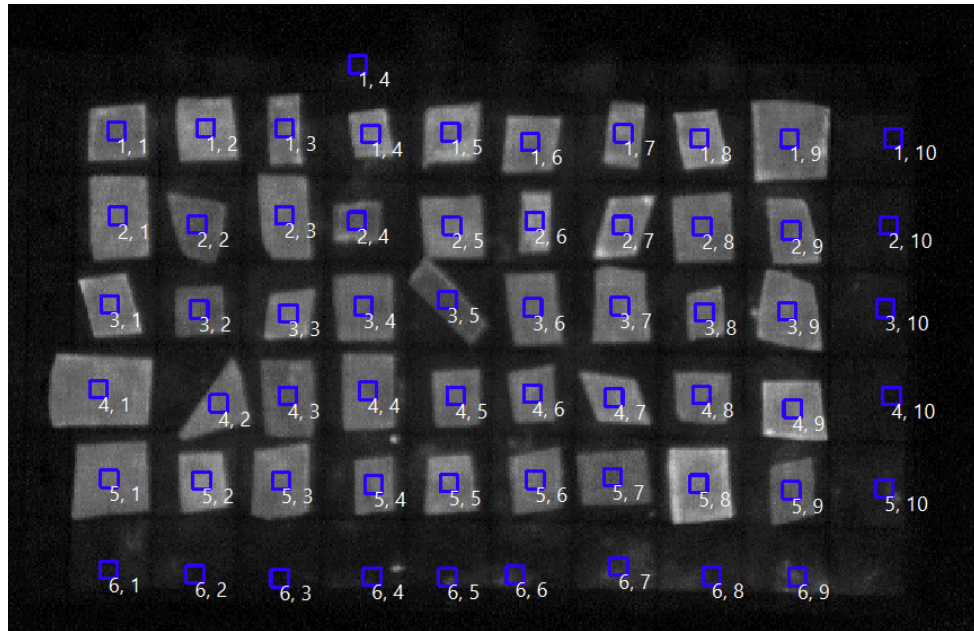
The image segmentation algorithm relies on a series of steps to identify likely ROIs in an array of leaf discs. Each step includes one or more parameters that may be tuned to optimize the performance of the image segmentation. **While we have attempted to provide reasonable default values for these parameters, individual circumstances may require adjustment to one or more parameters for best performance of the algorithm.** The parameters that can be adjusted are described below; the user is directed to the associated publication for a complete description of the image segmentation algorithm and associated parameters.

- **ROI size** (default 10 px): The length, in pixels, of one side of the square regions of interest.
- **Gaussian blur** (default 3 px): The size, in pixels of the Gaussian blurring kernel, used to reduce the influence of visual noise on subsequent processing steps.
- **Adaptive threshold neighbourhood size** (default 101 px): The size, in pixels, of the viewing region used during adaptive intensity thresholding.
- **Adaptive threshold constant offset** (default 2): A constant offset value subtracted from the intensity value of all pixels. Fine tuning this constant can help differentiate background from foreground.
- **Minimum leaf disc area** (default 100 px): The area, in pixels, of the minimum size required for a foreground object to be retained for further processing. This step can help prevent small pieces of debris from being identified as leaf discs.
- **Watershed segmentation size** (default 32 px): A characteristic size, in pixels, of leaf discs. This is used to separate multiple leaf discs when they are overlapping or touching.

Troubleshooting common problems

Background selected as ROIs

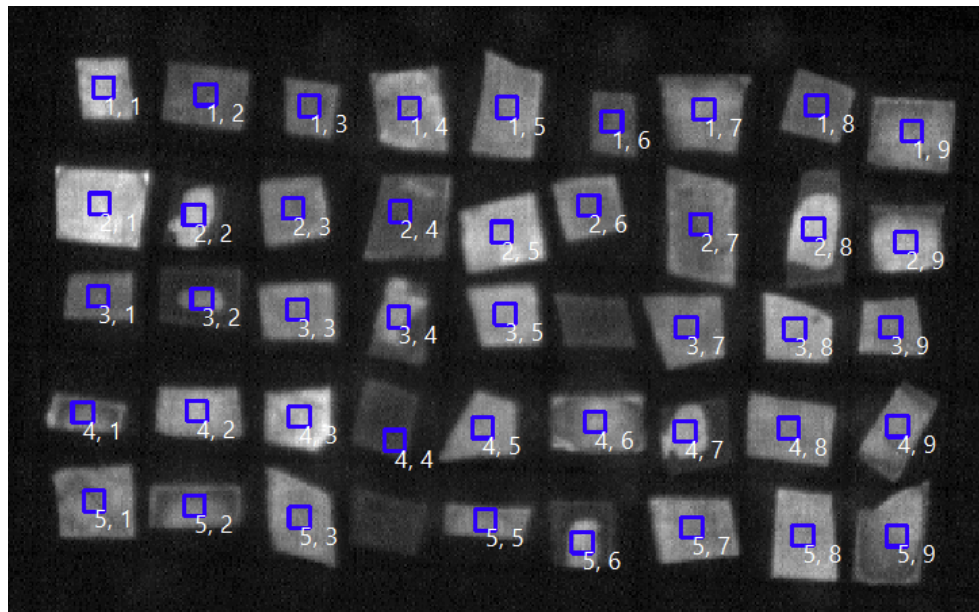
Example image:



Suggested action: decrease “adaptive threshold constant.”

Darker regions are not selected as ROIs

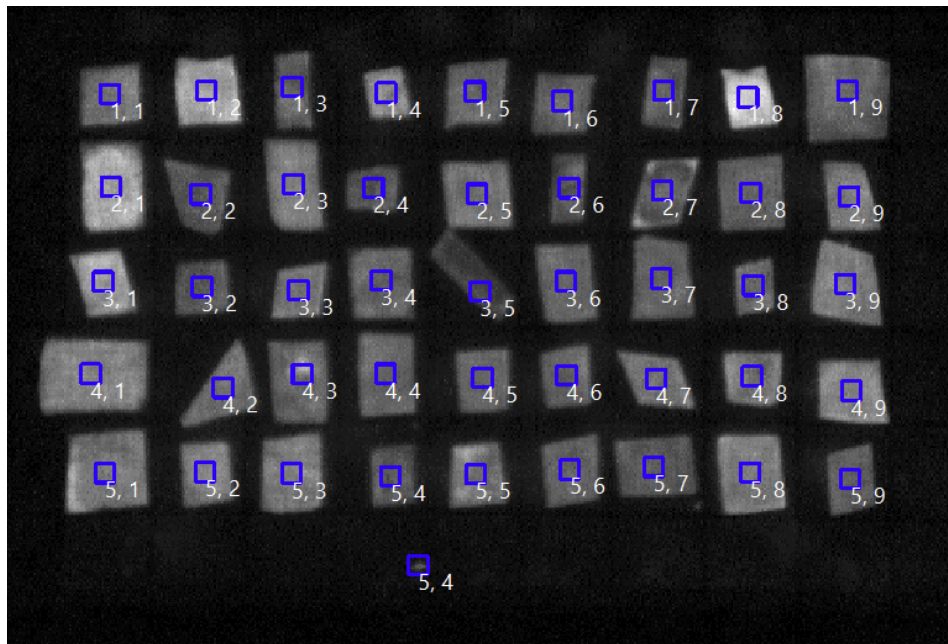
Example image:



Suggested action: increase “adaptive threshold constant.”

Small objects or debris are selected as ROIs

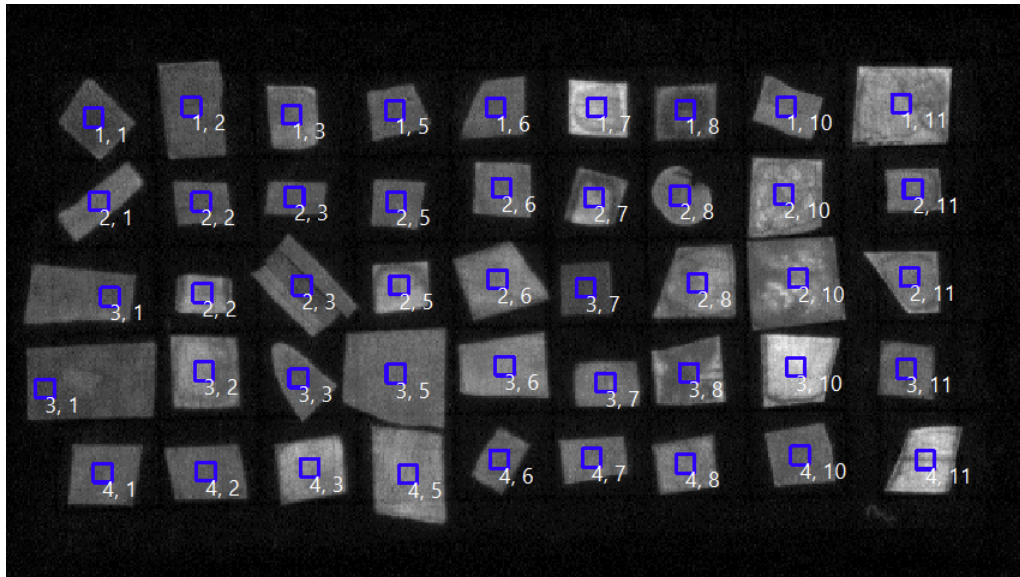
Example image:



Suggested action: increase “Minimum area,” or crop out background.

Incorrect row and column estimates or assignments

Example image:

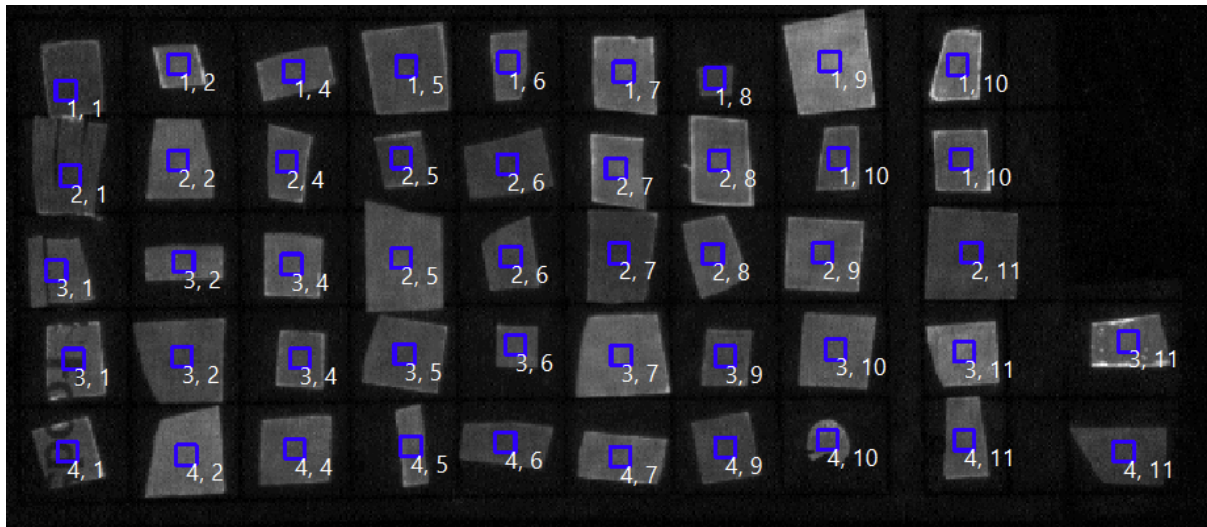


Suggested action: Adjust the placement of ROIs within the leaf discs such that ROIs are arranged in roughly linear rows and columns. Note in particular the placement of the ROI on the leaf disc located at row 4, column 1 (in the above image, this is incorrectly labelled as 3, 1). Moving this ROI to the right to be more in line with other ROIs in this column will improve row and column assignment.

Irregular or incomplete rectangular array

The use of incomplete or irregular arrays can result in poor performance of the row and column assignment algorithm.

Example image (note the incorrect assignment of row and column numbers):



Suggested action: complete the rectangular array by using “false ROIs”; these points can easily be removed from the datafile after export.

Example image (note row and column numbers are now correctly assigned):

